# KAYABA TRACKLESS AGV FIELD REPORT

3 NOVEMBER 2026

Between 13:47 and 15:28 WIB, vehicle MCQ4-4 made four autonomous trackless runs on route-itg-02 (map kyb-map-02 rev 3): out of the warehouse to the ITG station and back. **Only run 3 completed**, in 20 min 34 s, and it is the reference below. Run 1 faulted after three blocked-path retries. Run 2 stopped just off the line and could not resume. Run 4 lost three minutes to an obstacle the controller saw but the scanner did not, then was taken over manually. Runs 1 and 2 ran with faster corners and harder acceleration than runs 3 and 4. Run 3 was also the only run driven at **0.50 m/s** on straights; the others ran at 0.60 m/s.

## 1 · THE ROUTE: 32 STEPS

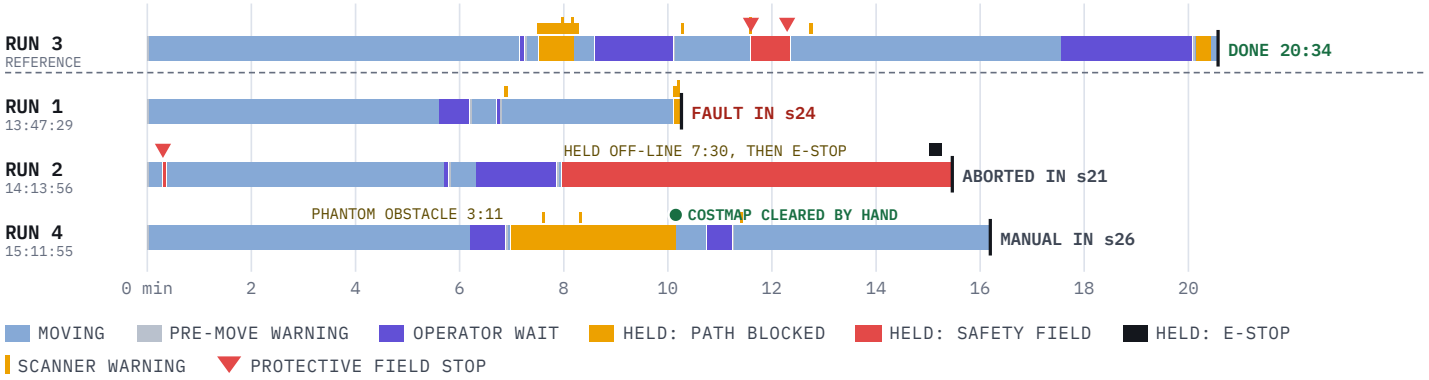
396 M · 14 STRAIGHTS, 15 ARCS, 3 WAITS

The executor sends the route to Nav2 one step at a time: each straight or arc is one goal, checked at its end before the next is sent. At a wait the vehicle stops until someone presses Continue. The route starts and ends at the same pose, (0, 0) facing east. The vehicle's log counts steps from zero, so its "step 11" is s12 here.

| | | | | |
|---|---|---|---|---|
| 1 | Out to ITG | s1-s10 | 199 m | From the start pose: 12 m north, 136.5 m west along the main aisle, 28 m north, S-bend into the ITG lane. |
| | Wait | s11 | | **ITG-RELEASE**: stops until Continue is pressed on the panel. |
| 2 | ITG maneuver | s12-s19 | 11 m | 180° turn at 1 m radius and a short jog to the attach point. |
| | Wait | s20 | | **ITG-ATTACH**: stops until Continue. This ends the outbound half (s1-s20). |
| 3 | Back to start | s21-s30 | 185 m | The same aisles in reverse: 28.5 m south, 116 m east, 13 m south, S-bend. |
| | Wait | s31 | | **ITG-RELEASE-HP**: stops until Continue. |
| 4 | Final approach | s32 | 1.2 m | Into the start pose. Inbound half is s21-s32. |

## 2 · TIMELINE

MINUTES SINCE START, RUNS ALIGNED AT ZERO




## 3 · RUNS

SPEEDS MEASURED FROM STEP TIMES · OUTBOUND S1-S20, INBOUND S21-S32

### RUN 1

FAULT

Fault in s24 · 13:47-13:57

| | |
|---|---|
| DURATION | 10 min 16 s |
| MOVING | 9 min 23 s |
| OUTBOUND | **6 min 4 s** s1-s20 |
| INBOUND | **3 min 19 s** to s24 |
| WAITS | 39 s |
| HELD | 8 s |
| STRAIGHTS | 0.60 m/s |
| CORNERS | 0.60 m/s |
| ACCEL | 0.3 m/s² |
| BRAKING | 0.5 m/s² |
| JERK LIMIT | none |
| IF BLOCKED | 3 retries, fault |

Path blocked 3 times on the 116 m straight s24; the executor then allowed only 3 retries.

### RUN 2

ABORTED

Aborted in s21 · 14:13-14:29

| | |
|---|---|
| DURATION | 15 min 28 s |
| MOVING | 6 min 9 s |
| OUTBOUND | **6 min 6 s** s1-s20 |
| INBOUND | **3 s** to s21 |
| WAITS | 1 min 39 s |
| HELD | 7 min 34 s |
| STRAIGHTS | 0.60 m/s |
| CORNERS | 0.60 m/s |
| ACCEL | 0.3 m/s² |
| BRAKING | 0.5 m/s² |
| JERK LIMIT | none |
| IF BLOCKED | 3 retries, fault |

After a hold on arc s21 it stood 0.13 m off the line, past the 0.10 m resume limit. Ended by E-stop and abort.

### RUN 3

REFERENCE

DONE

Complete · 14:45-15:05

| | |
|---|---|
| DURATION | 20 min 34 s |
| MOVING | 14 min 32 s |
| OUTBOUND | **7 min 45 s** s1-s20 |
| INBOUND | **6 min 46 s** |
| WAITS | 4 min 8 s |
| HELD | 1 min 46 s |
| STRAIGHTS | **0.50 m/s** |
| CORNERS | 0.30 m/s |
| ACCEL | 0.2 m/s² |
| BRAKING | 0.25 m/s² |
| JERK LIMIT | 0.4 m/s³ |
| IF BLOCKED | retry until clear |

Only complete run. Three holds (41, 46, 19 s), each cleared by itself.

### RUN 4

MANUAL

Manual in s26 · 15:11-15:28

| | |
|---|---|
| DURATION | 16 min 12 s |
| MOVING | 11 min 45 s |
| OUTBOUND | **6 min 50 s** s1-s20 |
| INBOUND | **4 min 55 s** to s26 |
| WAITS | 1 min 10 s |
| HELD | 3 min 11 s |
| STRAIGHTS | 0.59 m/s |
| CORNERS | 0.29 m/s |
| ACCEL | 0.2 m/s² |
| BRAKING | 0.25 m/s² |
| JERK LIMIT | 0.4 m/s³ |
| IF BLOCKED | retry until clear |

Held 3:11 at arc s12 by a phantom obstacle (72 retries); moved on after a manual costmap clear.

Source: field-logs/2026-11-03 (route_executor lines in ros-log/python3_*.log, events.jsonl, amr-logs/layer.log, route/route-itg-02_rev1.yaml). Speed is not logged. Straight speed = length / time moving on s5 + s7 (165 m); corner speed on the 2 m corners s2, s4, s6. "Moving" is the executor's moving state, less every retry the controller aborted again within 3 s: those never got the wheels going, so they count as held (1 s in run 1, 16 s in run 3, 35 s in run 4). What remains is within 2-4 s per run of the time Nav2 had a goal and the mux passed its command on. Acceleration, braking and jerk limit follow the a_max each base session logged at start (0.300 for runs 1-2, 0.200 for runs 3-4). Why run 3 drove 0.50 m/s is not recorded; the code on disk says 0.60.